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Wave simulation in 2D using FDTD with sparse tensor contractions

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Abstract

This thesis presents an alternative implementation and notation for the Finite-Differences Time-Domain (FDTD) method for the simulation of electromagnetic waves by translating the discretized derivatives as tensor operations, compared to explicit for-loops and array shifts used in standard approaches. The main goal of this thesis is a modular and user-friendly Python class to be used alongside Matplotlib for academic purposes, while providing a simple implementation and decent performance. The idea behind it is that the ever-evolving machine learning libraries may help provide a simple interface for highly optimized tensor operations. The code offers pre-programmed point-, line-, gaussian beam and phased-array sources together with the handling of dielectric and conductive materials, as well as absorbing boundary conditions. Further studies within this thesis include the comparison of analytical Fresnel coefficients and its numerical results obtained with the developed code. This was done by evaluating the power flux of incoming, reflected and refracted waves at planar dielectric interfaces.

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Chapter 1

Introduction

Since the development of computers, they have been extensively used to approximate solutions to differential equations with initial value problems describing the laws of nature. Electromagnetic waves are no exception. In 1966, mathematician Kane Yee published his seminal work [3], where he applied a leapfrog time integration on Maxwell's equations discretized as finite-difference equations. This algorithm is widely known as Finite Difference Time-Domain (FDTD), or, simply, Yee's algorithm. FDTD has proven to be extremely useful throughout the years due to its simplicity and effectiveness. Its application ranges from simple visualizations for didactic purposes, to the development of new waveguide technologies; for example: [4], [5], [6].

While the basic principles of FDTD over time, the used programming languages and implementation strategies have been adapted compared to Yee's original work. Hardware design or acceleration for certain operations may favour different implementations for higher performance, for example: vectorization, sparse linear algebra; while didactic may favour for-loops.

Parallel to the topic of wave simulation, the scene of artificial intelligence and machine learning has been growing quickly [7]. In 2015 the Python library Tensorflow was released, followed by PyTorch in 2016, providing easy access to machine learning models through the abstraction and optimization of tensor operations. This trend would only continue over the years. In February 2025 SciPy released its 1.15 version, which included tensor contraction operations for sparse arrays, which opened the opportunity for efficient tensors for FDTD.

Typically, numerical solutions of differential equations rely on the use of sparse linear algebra, in which the grid points of the typically two- or three-dimensional simulation are mapped to a one-dimensional vector to make use of vector-matrix multiplications. By using directly tensor operations, the reshaping of the arrays containing the grid point information can be easily skipped without increasing the time complexity of the linear transformations.

The following chapters will present a comprehensive introduction to FDTD and a detailed guide to a tensor-based implementation using numerical libraries in Python, including Numpy and SciPy.

Chapter 2

Electromagnetic theory and wave propagation

This work relies on the classic theory of electromagnetism as compiled and unified by James Clerk Maxwell. The first topic of this thesis shall therefore describe the emergence of electromagnetic waves and their propagation through space from Maxwell's equations.

2.1 Maxwell's Equations

Maxwell's equations are a set of four partial differential equations which describe the relations between electric and magnetic fields, and currents, as well as the reciprocal influence time-changing fields have on each other. These are

$$\nabla \cdot \mathbf{D} = \rho \quad (2.1)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (2.2)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (2.3)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (2.4)$$

All vector fields are implied to be functions of both position and time. For the sake of simplicity, we shall assume a linear, homogeneous and isotropic material, implying constant and scalar ε and μ . The constitutive relations are as such:

$$\mathbf{D} = \varepsilon \mathbf{E} \quad (2.5)$$

$$\mathbf{B} = \mu \mathbf{H} \quad (2.6)$$

$$\mathbf{J} = \sigma \mathbf{E} \text{ (Ohm's Law)} \quad (2.7)$$

In these equations:

$\mathbf{E}$ is the electric field

$\mathbf{D}$ is the electric flux density

$\mathbf{B}$ is the magnetic flux density

$\mathbf{H}$ is the magnetic field

ρ is the charge density

$\mathbf{J}$ is the current density

ε is the local electric permittivity

μ is the local magnetic permeability

σ is the local conductivity

2.2 The electromagnetic wave equations



2.2.1 Homogeneous equations

The approach to show that electromagnetic fields exhibit wavelike behaviour is by obtaining the wave equation from Maxwell's equations. The vectorial general homogeneous wave equation is defined as

$$\nabla^2 \Psi - \frac{1}{c^2} \frac{\partial^2 \Psi}{\partial t^2} = 0 \quad (2.8)$$

where ∇^2 is the vector Laplace operator.

Taking the curl of Faraday's Law (eq. 2.3) and substituting the constitutive equation 2.6 yields

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \quad (2.9)$$

← This is obtained directly with the general identity

$$\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F} \quad (2.10)$$

← Here, Schwarz's theorem has been implicitly invoked to switch the curl-operator and time-derivative. This is valid for stationary materials, since ε and μ remain as constants. Furthermore, since we are studying the propagation of electromagnetic fields, we shall consider a situation where no free charges nor free currents are present to act as local sources of electric and magnetic fields respectively, thus yielding $\nabla \cdot \mathbf{D} = 0$, and, via Ohm's law, $\nabla \cdot \mathbf{J} = 0$. Under these assumptions, by inserting Ampere-Maxwell's law 2.4 into 2.9, the wave equation or, rather, an attenuated wave equation is directly obtained:

$$\nabla^2 \mathbf{E} - \mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0 \quad (2.11)$$

An equivalent result can be **reached** the magnetic field by inserting Faraday's law (eq. 2.3) into the curl of Ampere-Maxwell's law (eq. 2.4), thus directly showing that electromagnetic fields naturally exhibit wavelike behaviour and are therefore capable of self-propagation solely by mutual induction.

2.2.2 Non-homogeneous equations

We shall now abstain from making **presuppositions** regarding the absence of free charges. In this case $\nabla \cdot \mathbf{E} \neq 0$, so applying 2.10 to 2.3 yields

~~$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu \frac{\partial}{\partial t} (\nabla \times \mathbf{H}) \quad (2.12)$$~~

~~Further resulting in a non-homogeneous wave equation for the electric field~~

$$\nabla^2 \mathbf{E} - \mu\sigma \frac{\partial \mathbf{E}}{\partial t} - \mu\varepsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} = \nabla(\nabla \cdot \mathbf{E}) \quad (2.13)$$

The derivation of the **magnetic** equation is slightly different since, contrary to the electric field, $\nabla \cdot \mathbf{H} = 0$. Using Ohm's Law (eq. 2.7) to substitute for $\mathbf{J}$ after taking the curl of Ampere-Maxwell's law (eq. 2.4), perhaps surprisingly, yields a homogeneous wave equation:

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H} = \sigma(\nabla \times \mathbf{E}) + \varepsilon \frac{\partial}{\partial t} (\nabla \times \mathbf{E})$$

$$\Rightarrow \nabla^2 \mathbf{H} - \mu\sigma \frac{\partial \mathbf{H}}{\partial t} - \mu\varepsilon \frac{\partial^2 \mathbf{H}}{\partial t^2} = 0 \quad (2.14)$$

At a first glance, one might assume an inconsistency between the equations governing electric and magnetic fields. Another question might arise: If antennas use currents to generate waves, why does the current density $\mathbf{J}$ not appear in these terms? This will be addressed in Section 2.3.

Solutions to the non-homogeneous wave equation

Similar to the solution of any linear operator, the solution to any non-homogenous wave equation is the superposition of a particular solution and the solution to the homogenous equation. Consider the following non-homogeneous equation:

$$\left(\nabla^2 - \alpha \frac{\partial}{\partial t} - \beta \frac{\partial^2}{\partial t^2} \right) \Psi(\mathbf{r}, t) = \xi(\mathbf{r}, t)$$

If $\varphi(\mathbf{r}, t)$ is a solution to the non-homogeneous equation and $\mathbf{h}(\mathbf{r}, t)$ is a solution to the homogeneous equation, $\varphi(\mathbf{r}, t) + \mathbf{h}(\mathbf{r}, t)$ is also a solution to the wave equation, because

$$\left(\nabla^2 - \alpha \frac{\partial}{\partial t} - \beta \frac{\partial^2}{\partial t^2} \right) (\varphi(\mathbf{r}, t) + \mathbf{h}(\mathbf{r}, t)) = \xi(\mathbf{r}, t)$$